K-Nearest Neighbors (KNN) - Complete Step-by-Step Solution

Question

Consider the following dataset and classify point P = (3,2) using KNN with k = 3.

Point	Feature1	Feature2	Class
A	1	1	Red
B	2	2	Red
C	3	3	Blue
D	6	5	Blue

Step 1: Distance Formula

Euclidean Distance = sqrt((x2-x1)^2 + (y2-y1)^2)

Step 2: Calculate Distances

Point	Calculation	Distance
A	√((3-1)^2 + (2-1)^2) = √5	2.24
B	√((3-2)^2 + (2-2)^2) = √1	1
C	√((3-3)^2 + (2-3)^2) = √1	1
D	√((3-6)^2 + (2-5)^2) = √18	4.24

Step 3: Select Nearest Neighbors (k=3)

The three closest points are: B, C, A

Step 4: Majority Voting

Neighbor	Class
B	Red
C	Blue
A	Red

Red = 2, Blue = 1

Final Answer

The predicted class of point P is Red.

Concept Explanation

KNN works by finding the nearest data points and using majority voting.

It assumes that similar points are close to each other in space.

Effect of k

Small k: sensitive to noise

Large k: smoother but less precise